Abstract

Can a shift in your brain's fat composition influence how a drug behaves? While traditional pharmacology primarily focuses on protein–ligand affinity, the lipophilic nature of cannabinoids means they must navigate a dense, heterogeneous lipid environment before reaching their receptors. In this study, we investigate how membrane composition influences the insertion and coupling of the cannabinoid agonist CP55,940. Using a combination of all–atom and coarse–grained molecular dynamics simulations, we characterize ligand behavior across multiple resolutions, from phase–defined bilayers to asymmetric synaptic models. Our results indicate that CP55,940 actively perturbs local membrane organization. In ordered bilayers, ligand insertion alters lipid packing, chain ordering, and interfacial hydration, whereas fluid membranes accommodate the ligand with minimal structural rearrangements. To explore the consequences of pathological lipidomic alterations, we examined a synaptic membrane depleted in ω–3 and ω–6 polyunsaturated fatty acids (PUFAs), inspired by trends reported in psychiatric disorders. Compared with the reference membrane, the PUFA–depleted system exhibits increased packing, higher lipid order, and reduced ligand insertion propensity. By connecting molecular simulations with biologically motivated lipid profiles, this work highlights membrane organization as a key factor in endocannabinoid signaling, illustrating how lipidomic variations shape the physical environment experienced by lipophilic drugs prior to receptor engagement.

Ret e voe din dibab ‘tre div vuhez,
En–dro d’ar familh, en–dro d’an enorioù.
Met n’eus ket a enor hep familh,
Setu perak emañ ar gerioù-mañ.

Papa, Maman, me ‘zo aet kuit
War hent ar studioù, pell diouzh an ti,
Pell diouzh ho karantez, pell diouzh ar c’hi,
Hag ar c’hazh sioul tal an tan.

Kavet em eus ur garantez,
Kollet em eus ur mignon,
Gouelañ a ran noz–ha–deiz,
Ne vo ken a «ken ar c’hentañ».

Soñjal a ran eus ma zud–kozh,
D’an amzer a dremen,
D’ar glav ha d’an amzer vat,
O c’hortoz an deiz, o tremen hep bezañ.

Remerciements

J’écris ces quelques mots pour remercier ceux qui ont côtoyé mon labeur, supporté mes humeurs et contribué à forger mon caractère.

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Contents

1. Introduction

1.1. Biological and pharmacological context

Cellular communication relies on molecular signals that must be detected and interpreted at the cell surface. In many cases, this process involves membrane receptors embedded in the plasma membrane, which translate extracellular or membrane–associated signals into intracellular responses. The membrane is therefore not only a physical boundary separating the cell from its environment, but also the cellular compartment in which many signaling events are initiated.

Traditional pharmacological frameworks implicitly assume that neurotransmitters diffuse through aqueous media before encountering their receptor [1]. This picture, while adequate for hydrophilic signaling molecules, becomes incomplete in the case of the endocannabinoid system (ECS) — a central regulator of neural activity, and a major pharmacological target in neurological and metabolic disorders. Endocannabinoids are lipid–derived molecules synthesized on demand from the hydrolysis of membrane phospholipids [2]. This biochemical origin confers lipophilic properties, meaning that these molecules preferentially partition into hydrophobic environments rather than the surrounding aqueous phase.

Among the main molecular targets of the ECS, the cannabinoid receptor type 1 (CB1) is one of the most abundantly expressed G protein–coupled receptors (GPCRs, a classic family of therapeutic targets that span the cell membrane to transmit extracellular signals inside) in the mammalian central nervous system. Structural and pharmacological evidence suggests that access to its orthosteric binding pocket may occur laterally from within the bilayer rather than directly from the extracellular bulk [3], [4]. This pathway is relevant not only for endogenous cannabinoids such as anandamide and 2-arachidonoylglycerol (2–AG), but also for exogenous ligands targeting CB1 [5]. These include phytocannabinoids such as Δ9–tetrahydrocannabinol (THC), as well as synthetic compounds developed for pharmacological purposes, among which CP55,940 and WIN55.212–2 are potent agonists (*i.e.* that bind to a receptor and activate it to produce a response), and rimonabant (SR141716A) a well–known inverse agonist (*i.e.* that binds to a receptor and reduces its activity).

The presynaptic membrane environment, which constitutes an essential component of endocannabinoid signaling, is enriched in sterols, phospholipids, and sphingolipids [6], [7]. Lipidomic analyses have revealed significant compositional alterations under pathological conditions, including depletion of ω–3 and ω–6 polyunsaturated fatty acids (PUFAs) in patients experiencing a first psychotic episode [8], [9]. From a biophysical standpoint, such compositional shifts modify membrane properties including acyl chain order, interfacial hydration, bilayer thickness and bending rigidity — parameters that directly influence ligand partitioning and mobility within the bilayer. Understanding how membrane composition modulates these ligand–membrane interactions therefore appears essential for describing the early physicochemical steps preceding CB1 engagement.

1.2. Lipid membranes as physicochemical systems

Lipid membranes are not homogeneous solvents but instead complex physicochemical systems, exhibiting rich phase behavior. This complexity is mainly brought by the diversity of their molecular constituents: headgroups, acyl chain saturation, and sterol content.

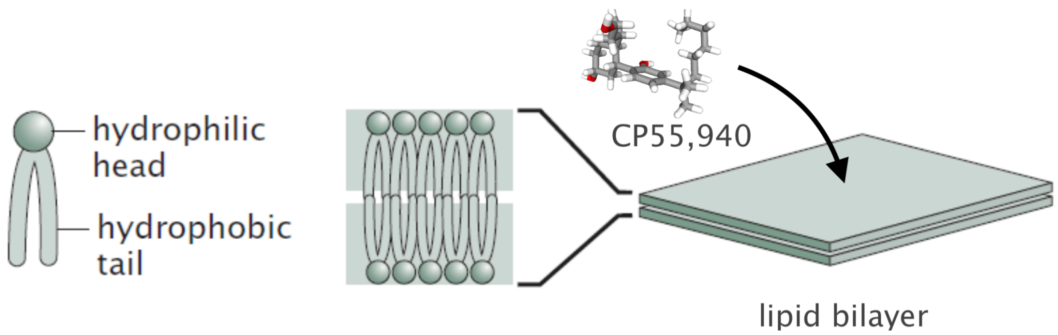

Figure 1 — Schematic representation of phospholipid amphiphilicity and bilayer organization. Hydrophilic headgroups orient toward the aqueous phase, whereas hydrophobic acyl chains assemble into the membrane core, leading to the spontaneous formation of lipid bilayers. Due to its lipophilic character, CP55,940 is expected to preferentially partition into this heterogeneous membrane environment rather than remain in the surrounding aqueous phase. Adapted from [10] .

The nomenclature of phospholipids reflects this structural organization. Species are typically designated by a four-letter code in which the first two letters indicate the two acyl chains, while the last two specify the headgroup. For instance, POPC is for *1-palmitoyl-2-oleoyl-sn-glycero-3-phosphocholine*, while SDPS denotes *1-stearoyl-2-docosahexaenoyl-sn-glycero-3-phospho-L-serine*.

Headgroups vary in size, charge distribution and hydrogen-bonding capability, thereby defining the interfacial character of the membrane [11]. These variations impose distinct geometric constraints at the molecular level, which propagate to the collective organization of the bilayer. *Phosphatidylcholines* (PC headgroup) exhibit an approximately cylindrical geometry favoring lamellar structures , whereas larger or smaller headgroups such as *phosphatidylethanolamine* (PE) introduce packing asymmetry and curvature stress.

Beneath this interfacial layer lies the hydrophobic core, mainly determined by the composition and chemical structure of acyl chains, also known as fatty acids [12]. Chain length and degree of unsaturation constitute the two primary levers to modulate the core. For example, increasing unsaturation enhances conformational flexibility, *i.e. cis* double bonds introduce kinks that disrupt tight packing, hence reduce chain order and increase membrane fluidity. Acyl chains are also commonly described using a compact notation. In a glycerophospholipid, the two chains are usually referred to as sn–1 and sn–2, corresponding to their position on the glycerol backbone. Another common notation specifies the number of carbon atoms and the number of double bonds in each chain. For example, DPPC contains two saturated 16–carbon chains and can therefore be written as 16:0/16:0, whereas a polyunsaturated chain containing 20 carbons and four double bonds would be denoted 20:4. The terms ω–3 and ω–6 further specify the position of the first double bond, counted from the end of the acyl chain; for instance, docosahexaenoic acid (22:6) is an ω–3 fatty acid since the first double bond is located between the third and fourth carbons when counting from that end.

Cell membranes also contain other essential lipids such as sterols and sphingomyelins. Among sterols, cholesterol plays a role in regulating membrane organization, fluidity and structural stability. Through their collective interactions, these different lipid species give rise to distinct membrane phase states. The phase diagram shown in Figure 2 corresponds to a model membrane composed of DPPC and cholesterol. The horizontal axis represents the cholesterol mole fraction, *i.e.* the proportion of cholesterol molecules relative to the total number of lipid molecules in the membrane mixture, while the vertical axis corresponds to temperature. Depending on these two parameters, the membrane may adopt different physical states, including the gel phase (L_β), the liquid–disordered phase (L_α, also commonly denoted L_d), and the liquid–ordered phase (L_o).

The gel phase (L_β) represents a highly ordered state. Within this regime, a tight packing of the lipid acyl chains is observed alongside a marked reduction in lateral mobility, which ultimately results in a rigid bilayer structure. A lower degree of chain ordering and enhanced conformational flexibility are exhibited by the liquid–disordered phase (L_α or L_d), further characterized by higher lateral diffusion rates. The liquid–ordered phase (L_o), typically found in cholesterol–rich membranes, combines features of both regimes.

Broad coexistence regions also appear in the diagram, such as $L_\alpha + L_o$ or $L_\beta + L_o$, indicating that different phases may coexist within the same membrane system depending on cholesterol concentration and temperature. Such coexistence reflects the heterogeneous nature of multicomponent membranes, in which local lipid organization and dynamics strongly depend on composition and thermodynamic conditions. Although phase equilibria are temperature–dependent, physiological conditions typically constrain this parameter within a narrow range.

In multicomponent systems, compositional heterogeneity may emerge at larger length scales. Lipid–lipid interactions between different species can drive partial lateral demixing [14], [15], leading to the formation of distinct domains within the same bilayer. Often discussed in the

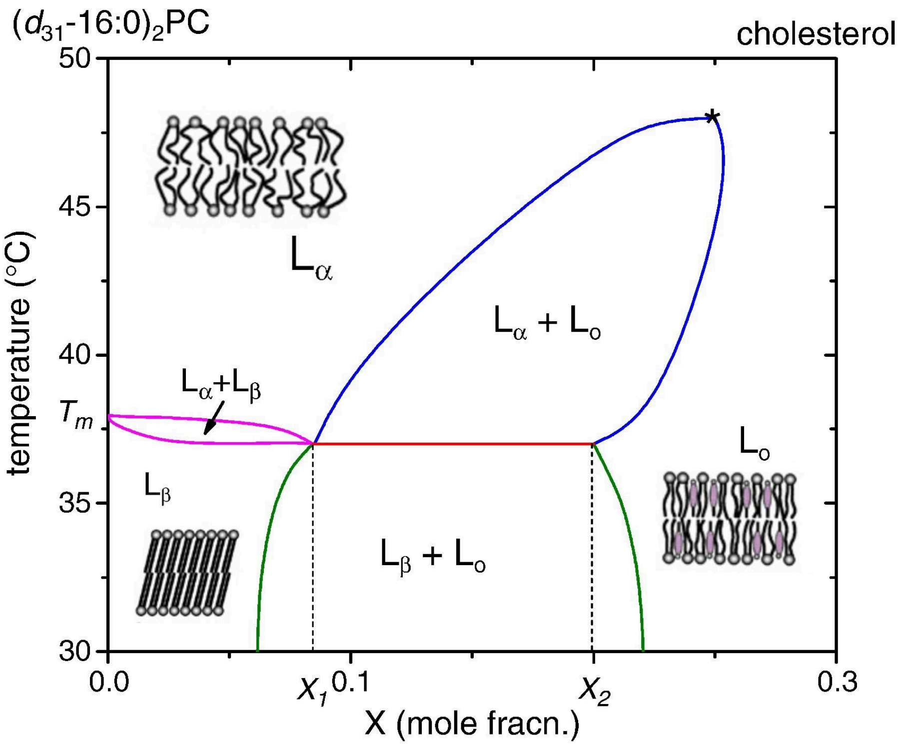

Figure 2 — Phase diagram of DPPC-cholesterol mixture as a function of temperature. Adapted from [13]. © Elsevier (2010), reproduced with permission.[1]

[1]License number n°*6241360511955*

Ligand	Class	logP
CP55,940	CB1 agonist	6.1
WIN55.212–2	CB1 agonist	4.4
SR141716A	CB1 antagonist / inverse agonist	6.5
Δ9-THC	Major phytocannabinoid	7.0
2–AG	Endocannabinoid	5.3

Table 1 — Octanol/water partition coefficients (logP) of representative cannabinoid ligands. Anything superior to 2 is considered lipophilic. Amphiphilic compounds lie between −2 and 2. Then, anything inferior to −2 is considered hydrophilic. It highlights lipophilic nature of CB1 ligands. For comparison, glucose exhibits a logP of approximately −2.6. All values were retrieved from PubChem.

context of lipid rafts, this mesoscale structure reflects the competition between entropy, which favors mixing, and enthalpy, which favors lipid–lipid associations, and results in regions exhibiting various mechanical properties.

1.3. Ligand–membrane coupling

While lipophilicity is often invoked to rationalize membrane affinity, a single partition coefficient (logP) cannot fully describe the complexity of ligand insertion into lipid bilayers, even though it is true that most CB1 ligands exhibit relatively high logP values as shown in Table 1. One reason is that the experimental determination of logP typically relies on octanol–water partitioning, a simplified system that lacks the headgroup region — which defines a chemically distinct interfacial environment.

Another limitation is that logP constitutes a purely thermodynamic descriptor and therefore provides little insight into the microscopic mechanism of insertion. It does not inform on whether a ligand preferentially enters through hydrophobic regions, whether it should reorient upon insertion, or how it organizes once embedded within the bilayer. Generally, it is advised to compute potential of mean force (PMFs) — that is, free–energy profiles evaluated along a chosen reaction coordinate — if one seeks better descriptions. PMFs provide access to the energetic cost associated with membrane entry, identify possible interfacial minima and reveal insertion barriers that could not be inferred from a global partition coefficient alone.

From this perspective, the membrane can be treated as an energy landscape experienced by the ligand. While such an approach is necessary to quantify insertion and identify potential barriers, it remains a reduced description of the interaction. The question arises: what is the physical nature of this interaction? A first level of consideration is steric. Much like inclusions in liquid crystals, an inserted ligand is expected to behave as a local defect, perturbing lipid packing and altering the orientational order of neighboring acyl chains. Additional mechanical effects may also emerge; the ligand may locally displace lipids upon insertion, generating curvature stress or elastic deformation of the surrounding.

Chemical and electrostatic contributions must also be considered. The membrane interface is characterized by a certain distribution of charges and dipoles, and screening effects are not necessarily uniform across the interfacial region. From a physical standpoint, the bilayer constitutes an electromagnetic environment, within which local interactions may further modulate ligand behavior.

Within this framework, CP55,940 emerges as a particularly relevant model ligand for CB1. Beyond its established pharmacological profile [16], [17], [18] as a potent CB1 agonist, it presents physicochemical characteristics that make it especially suitable for investigating ligand–-membrane coupling. It exhibits a relatively high partition coefficient, consistent with strong membrane affinity. Its molecular architecture combines an aromatic core and a hydrophobic aliphatic moiety, favoring insertion toward the inside of the bilayer, with additional polar functional groups capable of interacting with interfacial regions. Such amphiphilic balance makes it structurally compatible with prolonged residence within a bilayer while maintaining the ability to sample different insertion depths and orientations. It is therefore reasonable to treat CP55,940 as the molecular reference it has effectively become within the CB1 literature.

1.4. Objectives

The present work seeks to characterize how ligand–membrane coupling evolves as a function of lipid composition, in the specific context of CB1 ligands.

We aim, on the one hand, to quantify how ligand insertion perturbs the local structural organization of the bilayer. On the other hand, we examine how the membrane itself constrains and modulates ligand behavior, in a setting where the bilayer may constitute a privileged access pathway to CB1. This study is therefore deliberately positioned upstream of explicit receptor simulations.

To address these questions, we perform molecular dynamics simulations at different level of resolutions. First, relying on all–atom simulations, we aim at characterizing ligand insertion and local ordering at high structural details – being sensitive to fine structural rearrangements. These simulations are conducted in compositionally minimal membranes, within well–identified phase states, in order to isolate elementary physicochemical mechanisms.

We then extend the analysis to a coarse–grained level and to larger systems and longer timescales. This allows us to explore compositional effects at a broader scale and to assess whether local trends identified at the atomistic level persist in more complex membrane environments.

In particular, coarse–grained models make it possible to move beyond idealized binary mixtures and incorporate lipid compositions inspired by experimental lipidomics data. This is especially relevant for investigating variations in PUFAs content, whose imbalance has been reported in pathological contexts.

Accordingly, our main analytical tools consist of established membrane descriptors: membrane thickness, acyl chain order parameters, density profiles, PMFs or even hydration profiles. These indicators provide complementary structural and thermodynamic information and will be introduced and discussed systematically throughout the study. In line with this approach, we rely on a single ligand – CP55,940 – used as a reference in order to isolate membranotropic effects from ligand–specific variability.

Fundamentally, this work is guided by the idea that membrane composition must be regarded as an essential variable rather than a passive background. If the coupling is sensitive to lipid organization, then any variations in acyl chain saturation, sterol content, or PUFAs balance may alter drug behavior in ways that cannot be captured by protein affinity alone. The broader perspective of this research is to clarify to what extent the membrane itself participates in shaping pharmacological outcomes ; however, our current study remains focused on the mechanics of coupling.

2. Methods

2.1. All–atom molecular dynamics

2.1.1. Membrane compositions, system preparation

All–atom membrane systems were, for fluid systems, constructed using the CHARMM–GUI membrane builder [19]. Gel-like membranes were built by hand from a SLipids pre-equilibrated patch [20]. Symmetric bilayers composed of 100 lipids per leaflet were generated under periodic boundary conditions, resulting in lamellar multilayer systems separated by aqueous slabs.

To capture two completely distinct membrane physical states, DOPC and DPPC were chosen as reference phospholipids. At physiological temperature (37°C), DOPC 18:1/18:1 bilayers represent a fluid, liquid–disordered phase. Conversely, DPPC 16:0/16:0 membranes were simulated at 25°C — well below their main phase transition temperature of ~ 41°C. Setting the temperature here locks the membrane into a stable, gel–like phase, deliberately avoiding the strong fluctuations associated with the transition regime. Working with these two systems allows to study how ligand behavior changes in environments with different packing and chain ordering properties.

In addition to these reference membranes, compositional perturbations were introduced by partially substituting the host lipids with polyunsaturated species. In practice, 10 % of the lipids were replaced by either SAPC 18:0/20:4 (ω–6) or SDPC 18:0/22:6 (ω–3), leading to six distinct membrane compositions in total. These substitutions introduce highly unsaturated acyl chains into otherwise well-defined bilayers and allow us to probe the influence of polyunsaturated lipids on ligand–membrane interactions.

Due to the planar geometry of the bilayer and the use of periodic boundary conditions, the simulated systems naturally adopt a stacked lamellar configuration. To avoid interactions between periodic images of adjacent membranes, a sufficiently large aqueous slab was introduced, resulting in a hydration level of approximately 45 water molecules per lipid. Sodium and chloride ions were added to reproduce an isotonic salt concentration (0.15 M NaCl) representative of physiological conditions.

System preparation followed the standard equilibration procedure provided by CHARMM–GUI. After energy minimization, a restrained NVT equilibration was performed for 500 ps, followed by a restrained NPT equilibration of 1 ns using Berendsen barostat. The production equilibration runs were then performed for 150 ns for DOPC–based membranes and for 300 ns for DPPC–based ones, to allow the bilayers to reach equilibrium before further analysis. This was especially important for gel phase membranes, which exhibit slower lipid dynamics and therefore require longer equilibration times.

The ligand CP55,940 was introduced at concentrations reaching up to 10 mol% relative to the lipid content, after equilibration of membranes. To ease spontaneous insertion while avoiding artificial aggregation in the aqueous phase, ligand molecules were initially positioned close to the membrane surface. A pulling restraint along the membrane normal was applied during the early stages of equilibration in order to keep the ligands near the bilayer interface. After ligand insertion, an additional system relaxation step was performed. The systems were first subjected to energy minimization in order to remove possible steric clashes introduced during ligand placement. A short equilibration stage was then carried out before starting the production simulations.

2.1.2. Force field and simulation parameters

All–atom molecular dynamics simulations were performed using GROMACS 2018.2. Interactions were described using the CHARMM36m force field, which provides a well-established parametrization for lipid membranes and is widely used for phospholipid bilayer simulations [21].

Alternative membrane force fields exist, including so-called Berger model for lipids. However, CHARMM36m is known to correctly treat polar functional groups such as hydroxyl (-OH) moieties, which are relevant for describing ligand–membrane interactions involving hydrogen bonding at the membrane interface.

Ligand parameters for CP55,940 were generated using CGenFF, through the use of the ligand reader & modeler module of CHARMM-GUI [22]. Water molecules were represented using the TIP3P model, consistent with the CHARMM parametrization. This model is commonly employed in membrane simulations and remains computationally efficient compared to more complex water models due to its lower number of degrees of freedom.

The equations of motion were integrated using a 2 fs timestep during production runs, with all bonds involving hydrogen atoms constrained using the LINCS algorithm. Long-range electrostatic interactions were computed using the particle mesh Ewald (PME) method, with a real-space cutoff of 12Å. The same cutoff was applied to short-range van der Waals interactions.

Again, during production runs, temperature was controlled using a Nosé–Hoover thermostat, while pressure was maintained using a Parrinello–Rahman barostat with semi–isotropic pressure coupling, allowing independent fluctuations in the membrane plane and along the bilayer normal. The pressure coupling time constant was set to 5 ps.

2.1.3. Umbrella sampling and PMF calculations

To characterize ligand insertion beyond the simple partition coefficient, we considered potentials of mean force along a chosen reaction coordinate. In the present case, this coordinate was defined as the projection of the ligand center of mass along the membrane normal (z–axis), which provides a natural description of the insertion process.

From statistical mechanics, the PMF corresponds to a free–energy profile obtained by projecting the phase space distribution onto this reduced reaction coordinate. If x denotes a microscopic configuration of the system, and if $U(x)$ is its associated potential energy, the canonical probability distribution is given by the Boltzmann law,

$$P(x) = \frac{1}{Z} e^{-\beta U(x)}$$

where $\beta = 1/k_B T$ and Z is the partition function. Then, the probability associated with a given value of the reaction coordinate ξ is obtained by integrating out all remaining degrees of freedom, yielding,

$$P(\xi) = \frac{1}{Z} \int \mathrm{d}x e^{-\beta U(x)} \delta(\xi - \xi(x))$$

with δ the Dirac distribution. Finally, the corresponding free–energy profile is defined, up to an additive constant, as

$$F(\xi) = -k_B T \ln P(\xi) + \text{constant}$$

Broadly speaking, this quantity can be interpreted as an effective free–energy obtained after averaging over all microscopic configurations compatible with a given position of the ligand along the membrane normal.

In that sense, the free–energy profile can be reconstructed by varying the reaction coordinate of the ligand and, at each fixed position, estimating the associated probability distribution while allowing the ligand to explore the remaining degrees of freedom. Repeating this procedure over a series of overlapping windows gives access to the full profile. This procedure is known as umbrella sampling, and the ligand is maintained around a given value of the reaction coordinate by means of a harmonic biasing potential.

In practice, initial configurations for these windows were generated using *steered molecular dynamics* (SMD). After equilibration of the membrane–ligand system, the ligand was gradually pulled along the membrane normal, starting from the bulk aqueous phase toward the bilayer center.

Then, 35 configurations were extracted with a step size of $\sim$0.1 nm along this trajectory and then used as starting points for the umbrella sampling simulations, each of which was run for 100ns. The distributions obtained in each window were subsequently combined using the weighted histogram analysis method (WHAM) in order to reconstruct the unbiased PMF, since the harmonic potential introduces a bias.

In the present work, PMF calculations were restricted to DOPC membranes (64 lipids per leaflet) using GROMACS 2026.1. This choice was motivated by their fluid nature, which ensures sufficient molecular mobility and facilitates convergence of the free–energy profile along the chosen reaction coordinate. During steered molecular dynamics, the ligand was pulled along the membrane normal using a harmonic biasing potential, with a force constant of approximately 400kJ $\mathrm{mol}^{-1}\mathrm{nm}^{-2}$ and a constant pulling rate of $1 \times 10^{-4}\mathrm{nm\ ps}^{-1}$.

2.1.4. Observables

We characterized the structural and thermodynamic properties of the membrane using a set of standard observables, commonly employed in the field, probing the global state of the membrane and its local response to ligand insertion.

The first observable is the so-called area per lipid (APL), defined as the average surface occupied by a lipid in the membrane plane. APL is particularly sensitive to membrane composition and phase state, and provides a direct measure of lipid packing in a given leaflet. It also serves as a practical indicator of equilibration, as reference values for pure DOPC ($\sim$0.674 nm^2 at 30°C [23]) and DPPC membranes ($\sim$0.45 nm^2 at 20°C [24]) are well documented in the literature. The area per lipid was computed using a Voronoi tessellation of the membrane plane [25]. Each lipid was assigned an effective area corresponding to its Voronoi cell, constructed from the lateral positions of lipid headgroups. The APL was then obtained by averaging over all lipids within a leaflet.

The membrane thickness was measured as the average distance between the phosphate groups of the two opposing leaflets. Variations in thickness may reflect both compositional effects and asymmetry induced by local perturbations. Density profiles were evaluated by the use of histogram of atomic positions along a given axis, providing the average spatial distribution of molecular species and allowing the ligand position within the bilayer to be identified.

Hydration profiles were computed by counting the number of water molecules in the vicinity of the polar headgroup, using a distance-based cutoff, typically of the order of 3.5 Å — usually used for studying structuring waters from proteins [26]. Hydration provides a measure of solvent penetration at the membrane interface and is particularly relevant for characterizing this region, where polar interactions and hydrogen bonding play a significant role.

The orientational order of lipid acyl chains was also quantified, using the deuterium order parameter. This quantity is a direct measure of chain alignements with respect to the membrane normal and is defined as,

$$S_{\mathrm{CH}} = \frac{1}{2}\langle 3\cos^2\theta - 1\rangle$$

where θ is the angle between a given C-H bond vector and the normal, and where $\langle\cdot\rangle$ denotes a statistical average over time and lipids of the same species. The order parameter was computed for each carbon atom along the acyl chains using gorder [27], resulting in a position-dependent profile that reflects the variation of chain ordering from the headgroup region to the bilayer core. It is often understood that a higher value corresponds to a more rigid (more ordered) system, whereas values closer to zero indicate a more fluid system. However, this interpretation should be treated with caution, since the parameter only reflects an angular measurement.

Beyond these observables, molecular dynamics simulations can still be viewed as *in silico* experiments[2] and the trajectories provide additional qualitative insight into ligand behavior, such as insertion pathways or fine details of molecular arrangements.

2.2. Coarse–grained simulations

2.2.1. Martini 3 force field and ligand parametrization

Coarse–grained molecular dynamics was employed to study ligand–membrane coupling in lipid compositions of increased biological realism. While all–atom simulations provide detailed molecular information, they remain limited in terms of accessible system size and simulation timescale. These limitations become particularly important when considering lipidomic data, which may contain a large number of distinct lipid species, some of them representing only a few percent of the total membrane content. Moreover, several phenomena, including lateral lipid redistribution and partial demixing, occur on spatial and temporal scales that remain difficult to access using all–atom models.

[2] *Unfortunately, "simulation" has become increasingly misused to mean nothing more than "calculation."* – William L. Jorgensen [28]

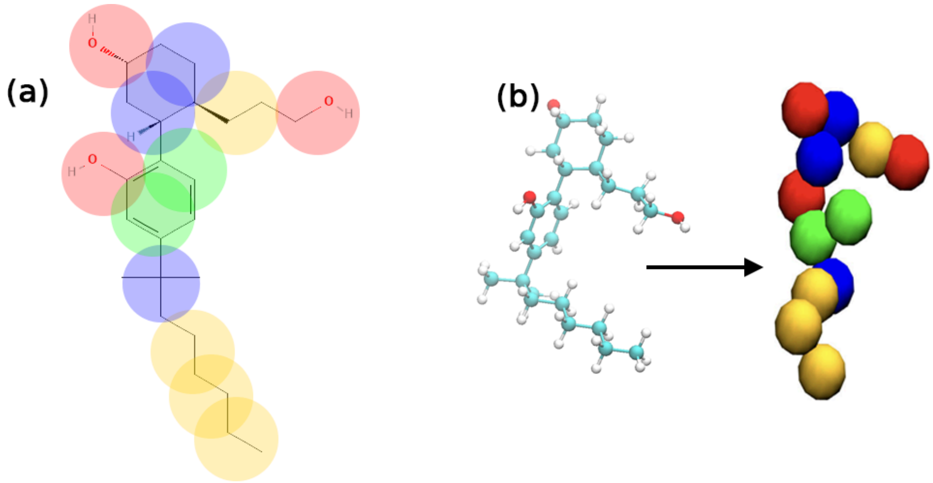

Figure 3 — MARTINI3 parametrization of CP55,940 obtained with Auto–MartiniM3. (a) Mapping strategy showing the correspondence between atomistic groups and coarse–grained beads. (b) Atomistic (left) and coarse–grained (right) representations of the ligand. Colors indicate MARTINI bead types: TP2d (red), SC1 (blue), TC5 (green) and TC1 (yellow).

To address these limitations, we employed the MARTINI3 (M3) [29] coarse–grained force field, in which several atoms are represented by a single interaction site. This reduction in resolution considerably decreases the computational cost of the simulations, allowing larger membrane patches and longer trajectories to be explored.

A large fraction of the lipid species required to reproduce the target lipidomic compositions is already available within the M3 ecosystem [30]. However, the parametrization of the ligand itself remains a critical step. In the present work, CP55,940 was coarse–grained using the Auto–MartiniM3 [31] tool, which provides an automated starting point for the construction of M3 models of small molecules. The availability of Auto–MartiniM3, together with the extensive documentation and validation studies accompanying M3, constituted an additional motivation for selecting this force field. Starting from the atomistic structure of CP55,940, Auto–MartiniM3 generated an initial mapping and set of interaction parameters, illustrated in Figure 3.

Because automated procedures may not fully reproduce the properties of a given molecule, additional validation steps were performed. In particular, PMF calculations were carried out for the coarse–grained ligand and systematically compared with the atomistic counterpart. The resulting model was judged sufficiently accurate to reproduce the main insertion features of CP55,940 but perfect quantitative agreement cannot be expected. The remaining discrepancies will be discussed in following sections.

2.2.2. Membrane composition

To move from simplified binary mixtures to more complex models, we constructed membrane compositions inspired by lipidomics analysis of the synaptic junction [32]. Rather than attempting to reproduce the complete lipidome, which contains a very large number of molecular species, we selected the fifteen most abundant lipid species identified in these datasets. Together, these species account for approximately 75% of the total membrane composition.

Particular attention was paid to preserving the asymmetric organization of the membrane. Lipids known to be enriched in the outer leaflet, such as sphingomyelins and ceramides, were assigned

Species	Structure	Outer	Inner	Total
CHOL	Cholesterol	923	878	1801
POPC	PC 16:0/18:1	609	60	669
DPPC	PC 16:0/16:0	344	34	378
PAPC	PC 16:0/20:4	118	14	132
SOPC	PC 18:0/18:1	97	12	109
SAPC	PC 18:0/20:4	97	12	109
PSPC	PC 16:0/18:0	74	9	83
PDPC	PC 16:0/22:6	51	7	58
SAPE	PE 18:0/20:4	86	340	426
SDPE	PE 18:0/22:6	54	216	270
SDPS	PS 18:0/22:6	0	424	424
SAPI	PI 18:0/20:4	0	293	293
ODLE	PE-O 18:1/18:1	0	195	195
SSM	SM 18:1/18:0	103	0	103
SCER	Cer 18:1/18:0	51	0	51
Total		**2607**	**2494**	**5101**

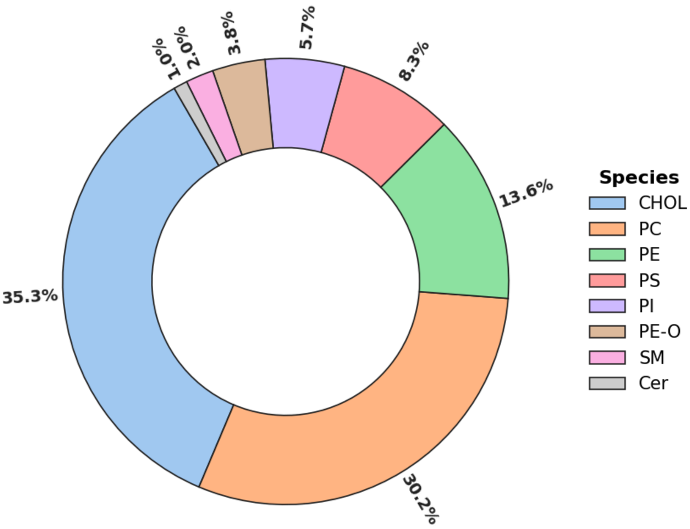

Table 2 — Lipid composition of the reference synaptic membrane model. The fifteen most abundant lipid species (75% of the experimental lipidome) were retained while preserving leaflet asymmetry. The pie chart shows the corresponding distribution of lipid classes.

to the extracellular side, whereas phosphatidylserines, phosphatidylinositols and plasmalogens were concentrated in the inner leaflet. The resulting membrane composition is reported in Table 2, with the corresponding distribution of lipid families summarized in a pie chart.

In addition to this reference membrane, a second composition was constructed to study the role of depletion in PUFAs. Experimental lipidomics studies have reported reduction in both ω–3 and ω–6 lipid species under pathological conditions [8], [9].

To reproduce this trend, we adopted a simplified depletion strategy. All polyunsaturated acyl chains, irrespective of their original degree of unsaturation, were systematically replaced by monounsaturated 18:1 chains. For instance, lipids containing arachidonic acid (20:4) or docosahexaenoic acid (22:6) were converted to the corresponding 18:1 analogues whenever available in the MARTINI 3 lipid library. Thus, SAPE (18:0/20:4) and SDPE (18:0/22:6) were replaced by SOPE (18:0/18:1), while SDPS (18:0/22:6) was replaced by SOPS (18:0/18:1).

In a few cases, no direct monounsaturated counterpart was available. The replacement was then performed using the closest lipid species within the M3 library while preserving the headgroup chemistry. For example, SAPI (18:0/20:4) was replaced by POPI (16:0/18:1), as no SOPI model was available. A similar approach was applied to the plasmalogen fraction.

This procedure preserves the overall distribution of lipid headgroups, and the asymmetry between leaflets while lowering to zero the abundance of ω–3 and ω–6 chains. The resulting membrane, reported in Table 3, should therefore be viewed not as a quantitative reconstruction of any specific pathological state, but more as a limiting model of PUFA depletion.

Species	Structure	Outer	Inner	Total
CHOL	Cholesterol	923	878	1801
POPC	PC 16:0/18:1	778	169	947
DPPC	PC 16:0/16:0	344	34	378
SOPC	PC 18:0/18:1	194	23	217
PSPC	PC 16:0/18:0	74	9	83
SOPE	PE 18:0/18:1	139	556	695
SOPS	PS 18:0/18:1	0	424	424
POPI	PI 16:0/18:1	0	293	293
DOLE	PE-O 18:1/18:1	0	195	195
SSM	SM 18:1/18:0	102	0	102
SCER	Cer 18:1/18:0	51	0	51
Total		**2604**	**2581**	**5185**

Table 3 — Simplified ω–3/ω–6 depleted membrane composition.

2.2.3. Simulation protocol

Coarse–grained membrane systems were constructed using the CHARMM–GUI MARTINI Maker [33]. After construction, the systems were subjected to the standard equilibration protocol generated by CHARMM–GUI, consisting of successive energy minimization and restrained equilibration steps.

Production simulations were subsequently performed using GROMACS 2026.1 in the NPT ensemble under periodic boundary conditions. Temperature was maintained at 310.15 K using the velocity–rescaling thermostat, while pressure was controlled semi–isotropically using the C–rescale barostat. Electrostatic interactions were treated using the reaction–field approach with a dielectric constant of 15. A timestep of 20 fs was employed throughout the production runs. All coarse–grained production trajectories were propagated for 2 µs.

CP55,940 molecules were then introduced at a concentration of 10 mol%, corresponding approximately to one ligand molecule for every ten lipids. Ligands were initially distributed on both sides of the membrane and the systems were subsequently equilibrated. Production simulations were then extended for an additional 2 µs in order to characterize ligand insertion and membrane reorganization processes.

2.3. Computational performances

Model	GROMACS	System size	Hardware	Performance	Compute time
All–atom	2018.2	$\sim$60k atoms	128 CPU cores	$\sim$75 ns/day	355k CPUh
MARTINI3	2026.1	$\sim$150k particles	NVIDIA V100	$\sim$1200 ns/day	1.3k GPUh

Table 4 — Computational resources used throughout this work. All–atom simulations were performed on the Mesocentre de Franche–Comté. MARTINI3 simulations were carried out on the Jean Zay national supercomputer through the ANR LADY.

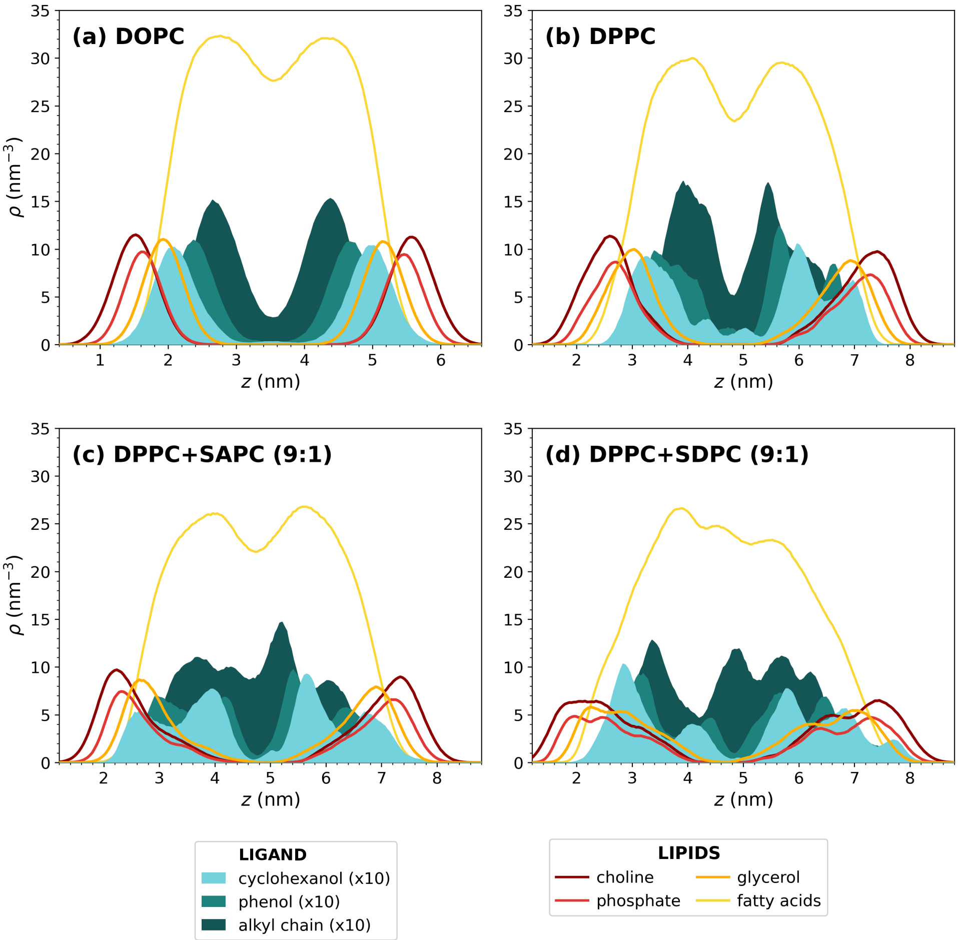

Figure 4 — Density profiles along the membrane normal for CP55,940 moieties and representative lipid groups in (a) DOPC, (b) DPPC, (c) DPPC+SAPC (9:1), (d) DPPC+SDPC (9:1) membranes. The cyclohexanol, phenol and alkyl–chain profiles correspond to the ligand, and corresponding densities are multiplied by 10 for better visualization.

3. Atomistic results: lipid dependent ligand-membrane coupling

3.1. Insertion profiles, orientational behaviors

In order to characterize the location of CP55,940 within the bilayer, density profiles were computed along the membrane normal for the different molecular groups — composing the ligand, and the lipids.

As shown in Figure 4, in pure DOPC layer (a), the ligand displays a preferential location within the interfacial region of the membrane, as its aromatic moieties remain predominantly localized near the glycerol backbone of the lipids. The aliphatic chain penetrates a bit deeper into the membrane – it tends to align with the surrounding chains, instead of completely folding on itself. This is expected for an amphiphilic insertion mode. We also observed a symmetric behavior upon enrichment with SAPC or SDPC lipids, suggesting that ligand localization is well preserved in these more disordered membranes.

A different behavior is reported in DPPC–containing systems (Figure 4 b–d). In contrast with the simple interfacial distribution in DOPC, the ligand density profiles exhibit multiple maxima extending deeper into the membrane. In particular, all the moieties display a bimodal distribution, with one population remaining close to glycerol and another penetrating significantly further into the core. This may be due to the coexistence of several metastable insertion states, possibly arising from slow relaxation dynamics imposed by more the ordered environment, leading to incomplete sampling on molecular dynamics timescales and contributing to the observed bimodal distribution.

Figure 5 illustrates the orientational distributions of CP55,940 aromatic core relative to the membrane normal. Fluid DOPC membrane (b) exhibits a relatively broad distribution centered around intermediate tilt angles (35–55°). In this configuration, the aromatic core remains close to the polar heads, allowing the hydroxyl groups to act as interfacial anchors in a polar region of the membrane.

Compared with DOPC, DPPC membranes (Figure 5 c–d) display less defined distributions, without a clearly preferred tilt angle, suggesting a more heterogeneous orientational landscape with multiple nearly equivalent configurations. This broader distribution may reflect a higher packing frustration and a more constrained interfacial environment, where the ligand orientation is less collectively organized and more sensitive to local membrane structure.

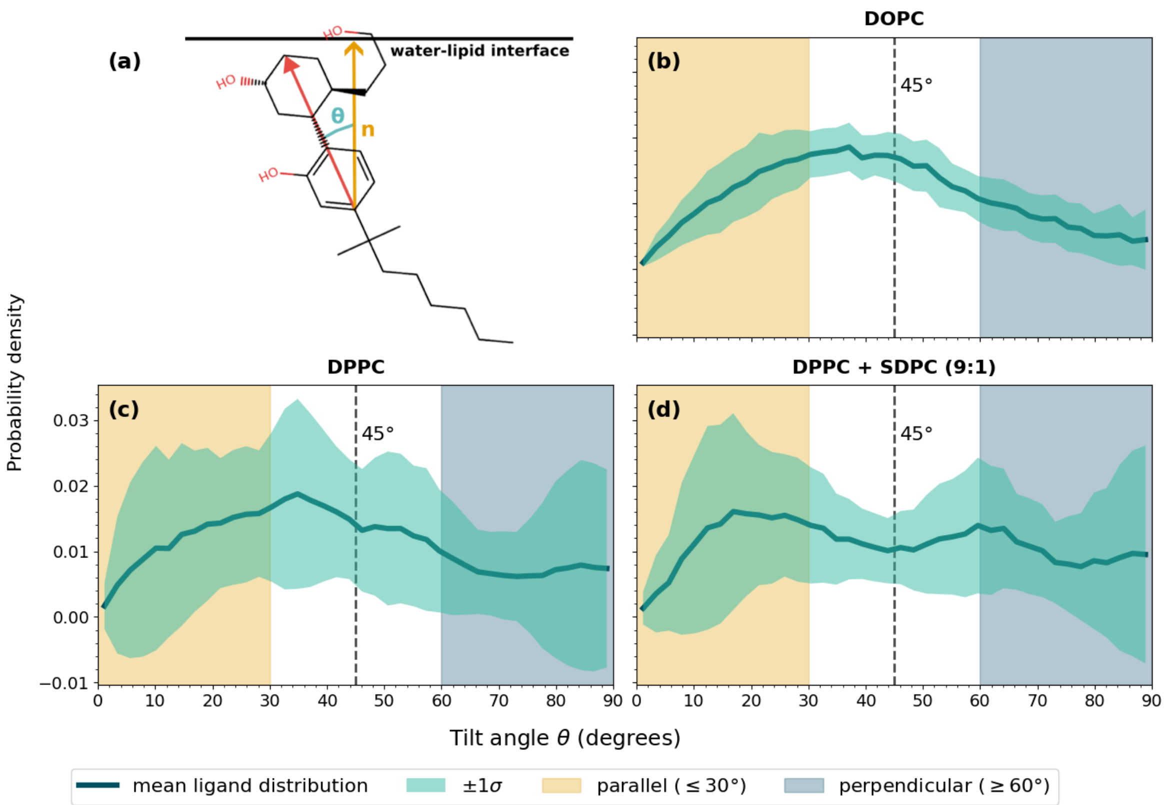

Figure 5 — Orientational distributions of the aromatic core of CP55,940 relative to the membrane normal in (b) DOPC, (c) DPPC and (d) DPPC+SDPC (9:1) membranes. (a) Schematic representation of the tilt-angle definition.

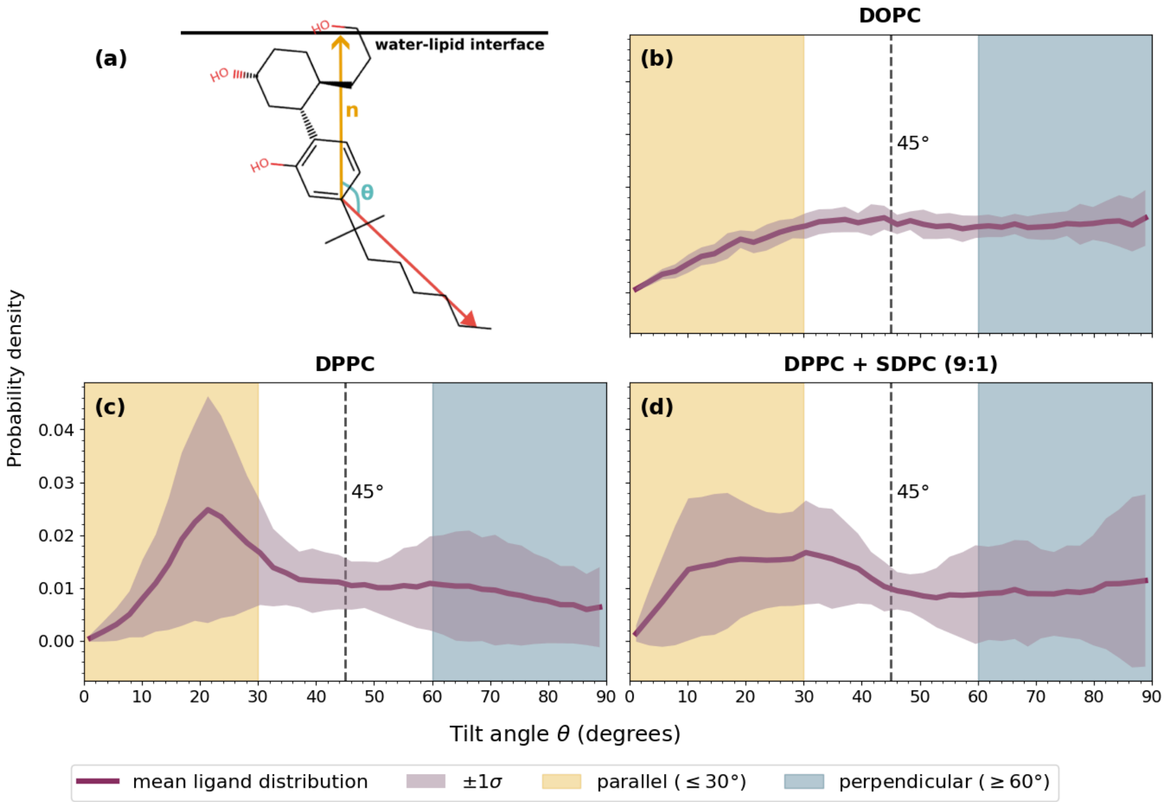

Figure 6 — Orientational distributions of the alkyl chain of CP55,940 relative to the membrane normal in (b) DOPC, (c) DPPC and (d) DPPC+SDPC (9:1) membranes. (a) Schematic representation of the tilt-angle definition.

This interpretation is supported by Figure 6, where we report orientational distributions of the alkyl chain of the ligand. The DOPC membrane (b) shows an almost isotropic distribution, with no preferential tilt angle over a wide angular range. This behavior suggests a highly fluid and permissive environment, where the ligand retains orientational freedom and can explore multiple configurations. In contrast, the DPPC system (c) display a marked low-angle preference (20 °), indicative of a more tightly packed surrounding that constrains the alkyl chain. An intermediate behavior is observed for DPPC+SDPC (d), supporting the idea that introducing PUFA increases lipid disorder progressively and restores flexibility.

Taken together, these observations suggest that the insertion behavior of CP55,940 results from a balance between chemical affinity for the membrane interface and steric constraints imposed by the lipid matrix. In fluid condition, the ligand appears to be primarily guided by favorable chemical interactions, as the hydroxyl groups remain anchored near the hydrated interfacial region while the rest of the molecule retains conformational freedom. This picture is consistent with previous experimental NMR results obtained for CP55,940 in POPC membranes [17], a fluid phosphatidylcholine bilayer close to DOPC, where the ligand was also reported to adopt an interfacial insertion mode. It contrasts with ordered membranes, where DPPC imposes stronger packing constraints, leading to broader insertion distributions and less well–defined orientational preferences. In these systems, local membrane organization likely competes with the intrinsic amphiphilic character of the ligand in determining its insertion mode. The introduction

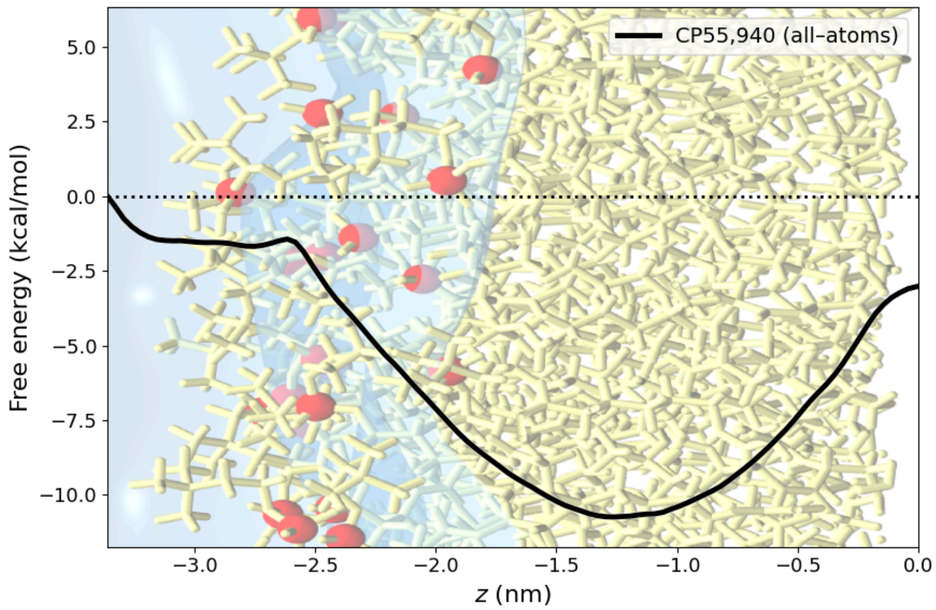

Figure 7 — Potential of mean force computed for a CP55,940 ligand approaching a DOPC membrane in all–atom. An indicative illustration is set in the background to visualize the leaflet.

of unsaturations progressively relaxes these constraints, partially restoring the configurational flexibility and interfacial behavior.

3.2. Free energy profiles (PMF)

Having established from insertion profiles that CP55,940 adopts preferentially an interfacial insertion mode in fluid DOPC membranes, we next quantified the energetic cost associated to it. The free–energy profile shown in Figure 7 was reconstructed by taking the aqueous phase as the reference state.

Starting from bulk water, the PMF rapidly decreases as the ligand approaches the interface, indicating a thermodynamic driving force for membrane insertion, confirming that the membrane constitutes a favorable environment for the ligand. Interestingly, a very small energy barrier ($\sim$0.25 kcal·mol^{-1}, that is $\sim$0.4 kT at physiological temperature) is observed close to the water-lipid interface. Although small compared to thermal fluctuation, it likely reflects the energetic cost associated with crossing the phosphate headgroup region and locally perturbing lipid packing in order to access the hydrophobic core of the bilayer.

Once this barrier is crossed, the free–energy decreases toward a pronounced minimum of ~ -11 kcal·mol^{-1} located inside the membrane, in good agreement with the density profiles and orientational analyses. Finally, the PMF rises again when approaching the bilayer center, usually depleted in lipids. In practice, this free–energy increase should act as a barrier against spontaneous flip–flop events on accessible simulation timescales.

3.3. Structural and mechanical response

We next examined how the membrane itself responds to ligand incorporation. To this end, we analyzed the area per lipid, the deuterium order parameter and the local hydration environment of the phosphate, which together should provide complementary information on membrane

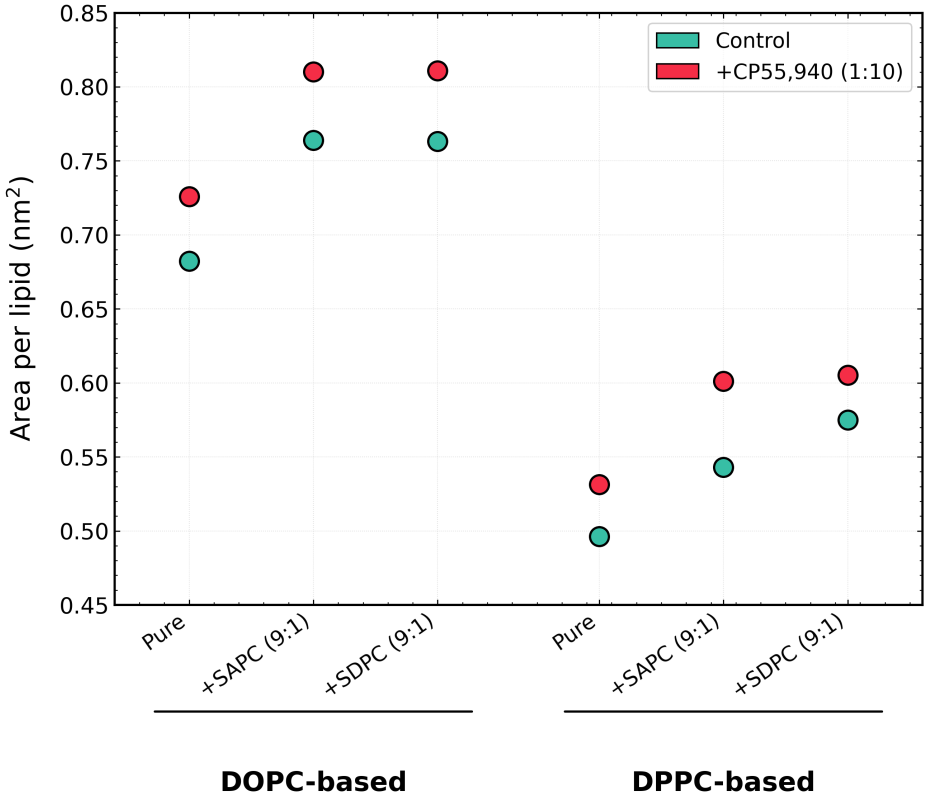

Figure 8 — Area per lipid (APL) measured for DOPC–based and DPPC–based membranes in the absence (control) and presence of CP55,940. For mixed systems, the reported values correspond to the average area per host lipid within the membrane plane.

packing, ordering and interfacial organization. In mixed membranes, order parameter analyses were restricted to the host lipids (DOPC or DPPC). Indeed, the relatively small number of SAPC or SDPC molecules in the simulated patches did not allow statistically robust profiles to be extracted for the polyunsaturated species themselves.

The first important thing to note is that the control systems reproduce the expected physical trends associated with membrane composition and phase state, and are coherent with data from the literature [23], [24]. As shown in Figure 8 and Figure 10, DPPC–based membranes display smaller APL values together with substantially higher order parameters than DOPC systems (Figure 9), reflecting the tightly packed and more ordered nature of the gel phase. DPPC membranes enriched with SAPC or SDPC display though slightly higher order parameters than pure DPPC. At first sight, this may appear counterintuitive, since the introduction of unsaturated chains is generally associated with increased disorder. However, this observation should be interpreted carefully. In the gel phase, DPPC acyl chains are strongly tilted with respect to the membrane normal, while remaining very straight, as suggested by the flat–looking curve. Still, since the order parameter only probes the angle formed by C–H bonds relative to the normal, such collective tilt can artificially reduce the measured S_{CH} despite the membrane remaining rigid. The introduction of polyunsaturated lipids likely shifts the membrane away from the gel organization toward a more fluid but still ordered state, possibly closer to a liquid–ordered regime. In this configuration, the average chain tilt relative to the membrane normal is

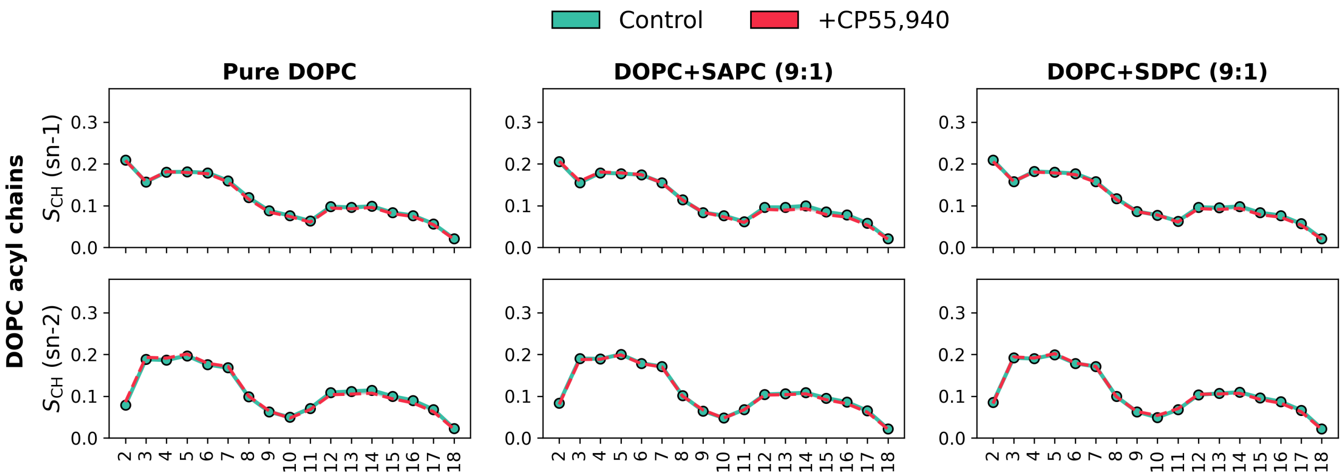

Figure 9 — Deuterium order parameter profiles (S_{CH}) computed for DOPC acyl chains in pure DOPC, DOPC+SAPC (9:1) and DOPC+SDPC (9:1) membranes, in the absence (control) and presence of CP55,940. The upper and lower panels correspond respectively to the sn–1 and sn–2 chains of DOPC. Order parameters were computed only for the host DOPC lipids.

reduced, which may contribute to the larger apparent order parameters observed here. We also noted that the profiles become more rounded and less flat, suggesting that part of the extreme packing characteristic of the gel phase is progressively lost, together with some restored fluidity. Conversely, DOPC bilayers exhibit larger APL values and lower order parameters, in agreement with their liquid–disordered nature at physiological temperature. The addition of unsaturations further shifts the systems toward more disordered configurations. Overall, these observations are consistent with the expected hierarchy between the different membrane compositions and therefore support the physical coherence of the simulated systems.

In this background, the effect of CP55,940 appears to be dependent on the initial membrane organization. In all systems, ligand insertion induces an increase in area per lipid, indicating a global lateral expansion — depacking — of the bilayer. This effect remains moderate in fluid DOPC membranes but becomes more pronounced in DPPC systems. From a physical perspective, this behavior is compatible with the ligand acting as a local inclusion within the membrane, perturbing the packing and generating additional free volume around the insertion region. This expansion is not systematically accompanied by a decrease in chain ordering. In DOPC membranes, the deuterium parameter profiles remain almost unchanged upon ligand insertion, suggesting that the fluid bilayer can accommodate the perturbation without major structural reorganization. By contrast, DPPC–based membranes display a noticeable increase in chain ordering in the presence of the ligand, which we interpret as a trace of packing rearrangement around the ligand, where the membrane reorganizes in order to compensate for the perturbation introduced by insertion. The same behavior is observed upon insertion of SAPC and SDPC, and these effects are cumulative.

Hydration profiles further support the observed differences between the various membrane compositions. As represented in Figure 11, DOPC–based membranes exhibit systematically larger hydration levels around the phosphate group than DPPC systems. The introduction of polyunsaturated lipids increases hydration in both membrane families, an effect that is particularly pronounced in DPPC systems. This behavior is coherent with the increase in area per lipid

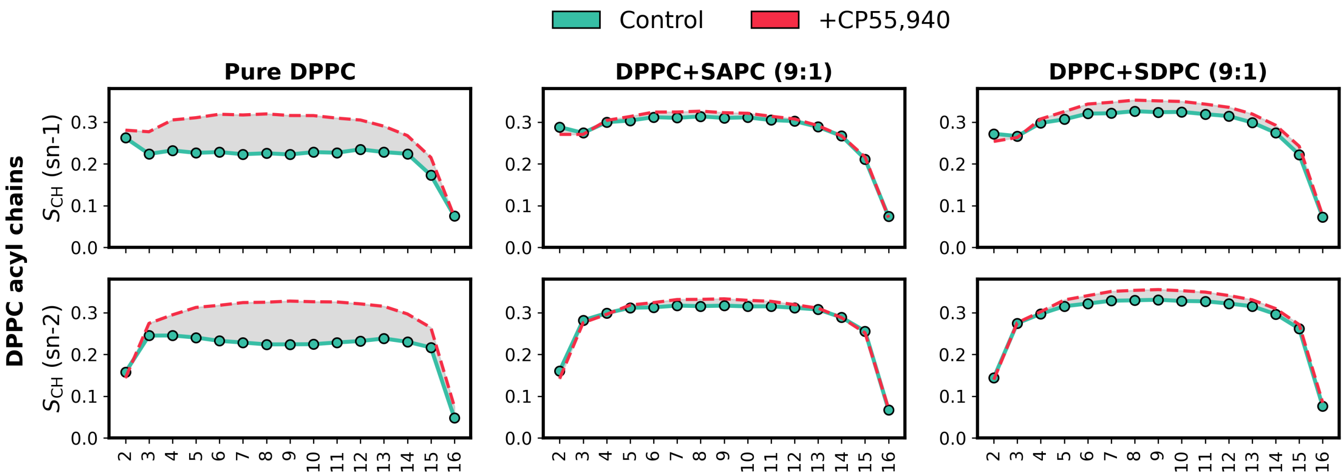

Figure 10 — Deuterium order parameter profiles (S_{CH}) computed for DPPC acyl chains in pure DPPC, DPPC+SAPC (9:1) and DPPC+SDPC (9:1) membranes, for both control systems and membranes containing CP55,940. The upper and lower panels correspond respectively to the sn–1 and sn–2 chains of DPPC. Order parameters were evaluated only on the host DPPC lipids.

and the partial loss of the highly packed gel organization discussed previously. The insertion of unsaturated chains likely introduces local packing defects and additional free volume, thereby facilitating water penetration toward the interfacial region.

The presence of CP55,940 also shifts the distributions toward larger hydration values. This effect remains moderate in fluid DOPC membranes but becomes much more visible in DPPC systems. Altogether, these observations support the idea that both polyunsaturated lipids and ligand insertion contribute to relaxing the compact organization of ordered membranes and promote a more hydrated interfacial environment.

For the sake of clarity, all hydration distributions were normalized by the number of host lipids (DOPC or DPPC), allowing direct comparison between the different membrane compositions despite the presence of mixed systems.

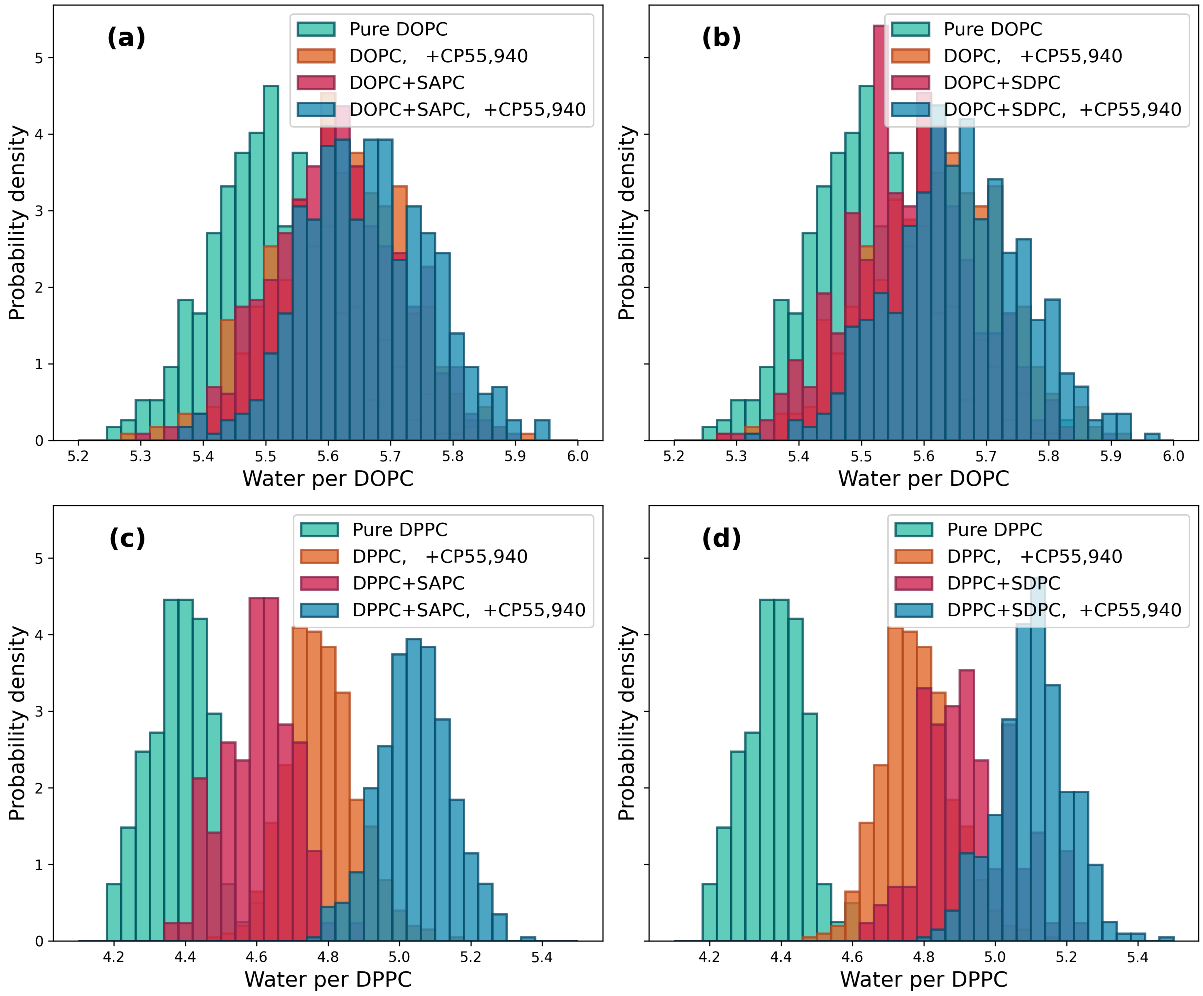

Figure 11 — Hydration distributions computed around the phosphate moieties of the host lipids for (a) DOPC and DOPC+SAPC (9:1) membranes, (b) DOPC and DOPC+SDPC (9:1) membranes, (c) DPPC and DPPC+SAPC (9:1) membranes, and (d) DPPC and DPPC+SDPC (9:1) membranes, in the absence and presence of CP55,940. The distributions are normalized by the number of host lipids (DOPC or DPPC).

Model	APL (nm²)	Thickness (nm)	S_{CH}
All-atom	0.68 ± 0.03	3.86 ± 1.59	0.113 ± 0.156
MARTINI3	0.65 ± 0.03	3.86 ± 1.23	0.333 ± 0.216

Table 5 — Structural membrane properties obtained from all-atom and MARTINI3 simulations of DOPC bilayers. Uncertainties correspond to three standard deviations (3σ).

4. Extension to lipidomic context: coarse–grained results

4.1. Multiscale consistency

To study the consistency between all–atom and coarse–grained descriptions, a DOPC membrane was simulated using both representations and compared. The coarse–grained model contained 128 lipids per leaflet, whereas the atomistic models correspond to those previously studied (*i.e.* 100 lipid per leaflet for structural properties, 64 per leaflet for PMF). The structural properties of the membranes are reported in Table 5.

The area per lipid obtained with the all–atom model is in excellent agreement with the M3 value, the difference remaining within the three–sigma uncertainty interval. Similarly, both models predict an identical membrane thickness of 3.86 nm. These results indicate that the coarse–grained representation reproduces nicely the global structural organization of the membrane.

Larger discrepancies are observed for the deuterium order parameter, but a quantitative comparison should be interpreted with caution. In atomistic simulations, the deuterium order parameter is defined from the orientation of C–H bonds relative to the membrane normal. Since hydrogen atoms are not explicitly represented in M3, the corresponding quantity is estimated from C–C bonds, implying that the two observables are not strictly equivalent. Still, both models remain consistent with a fluid state, although the coarse–grained representation appears somewhat more ordered. This behavior may arise from the reduced number of degrees of freedom associated with coarse–graining, which might restrict local conformational fluctuations.

A more stringent test is provided by free–energy profiles associated with membrane insertion of CP55,940. As shown in Figure 12, both models predict favorable membrane partitioning. In particular, the minima are located at comparable depths within the bilayer and exhibit similar amplitudes, close to -11 kcal·mol^{-1}. Differences remain in the detailed shape of the PMF curve. The atomistic profile exhibits a significant free–energy increase toward the bilayer center, which is broader within the M3 model. As a consequence, the coarse–grained ligand is expected to explore a larger fraction of the bilayer interior and may diffuse more readily once inserted, allowing for flip–flop events to occur. This likely suggests a stronger effective hydrophobicity of the coarse–grained representation. However, such an interpretation should remain cautious: if it is known that M3 force field generally produce faster diffusion dynamics than atomistic models, the apparent confinement observed in the atomistic PMF may also partly result from incomplete sampling of slow rearrangements. Therefore, the observed differences cannot be unambiguously attributed to the ligand parametrization alone.

Overall, the M3 model reproduces the essential structural characteristics of the membrane and preserves the main thermodynamic features of CP55,940 insertion. Although quantitative

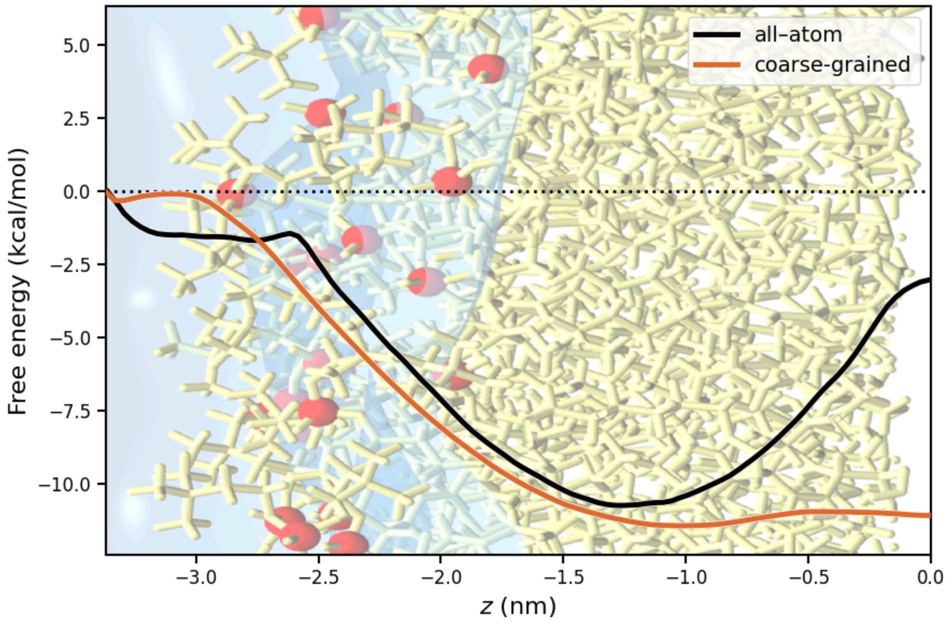

Figure 12 — Potential of mean force associated with CP55,940 insertion into a DOPC bilayer obtained at atomistic and coarse–grained resolutions.

differences remain in orientational descriptors and in the fine structure of the PMF, the overall insertion mechanism appears conserved across resolutions. These results support the use of the coarse–grained model for exploring larger membrane systems.

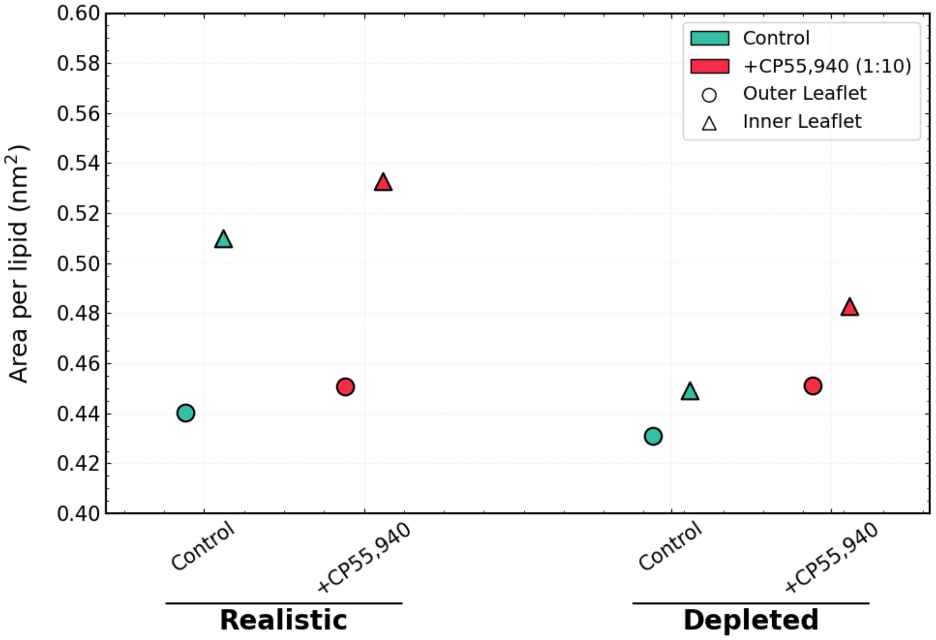

Figure 13 — Area per lipid (APL) of the realistic and PUFA–depleted membranes in the absence and presence of CP55,940. The reported values correspond to the average area per lipid of each leaflet, considering all lipid species. Circles denote the outer leaflet and triangles the inner leaflet.

4.2. Realistic membrane compositions

The realistic and PUFA–depleted compositions were then compared at the coarse–grained level in order to assess how lipidomic changes affect membrane organization and ligand insertion. As shown in Figure 13, the realistic membrane exhibits a marked leaflet asymmetry, with a larger APL in the inner leaflet than in the outer leaflet. This is consistent with the asymmetric composition of the model, since the inner leaflet contains a larger fraction of PE, PS, PI and polyunsaturated lipid species. In contrast, PUFA depletion strongly reduces this asymmetry and decreases the APL, especially in the inner leaflet. This indicates that replacing polyunsaturated chains by monounsaturated analogues leads to a more compact membrane organization.

The addition of CP55,940 increases the APL in both membrane models, indicating that the depacking effect observed at the atomistic level is preserved in the larger coarse–grained systems. This effect is particularly clear in the realistic membrane, where both leaflets expand upon ligand addition. In the PUFA–depleted membrane, CP55,940 also increases the APL, suggesting that the ligand still acts as a membrane inclusion that perturbs lateral packing, even in a more ordered lipid environment.

This interpretation is further supported by chain order parameter values reported in Table 6. The PUFA–depleted membrane displays a higher average order parameter than the realistic membrane, increasing from 0.3742 ± 0.0005 to 0.4994 ± 0.0002 in the control systems. This confirms that PUFA depletion produces a more ordered and tightly packed bilayer. Upon addition of CP55,940, the order parameter decreases in both membranes, from 0.3742 to 0.3576 in the realistic system and from 0.4994 to 0.4558 in the PUFA–depleted system. Together with the APL increase, this shows that CP55,940 tends to disorder the membrane and partially counteracts the compaction induced by PUFA depletion.

Finally, ligand insertion was quantified by counting the number of CP55,940 molecules effectively inserted into the membrane after equilibration. In the realistic membrane, 458 out of 500 ligands were found inserted, whereas only 360 out of 500 ligands were inserted in the PUFA–depleted membrane. Although this analysis remains qualitative, because some ligands can aggregate before insertion, it suggests that the realistic PUFA–containing membrane provides a more permissive environment for CP55,940 uptake. Conversely, the more ordered PUFA–depleted membrane appears less favorable to ligand insertion on the simulated timescale.

Overall, these coarse–grained simulations extend the atomistic observations to larger and more complex membrane models. PUFA depletion induces membrane compaction and increased chain ordering, whereas CP55,940 promotes lateral expansion and decreases lipid order. The lower insertion fraction observed in the depleted membrane further suggests that lipidomic

System	Control	+CP55,940
Realistic	0.3742 ± 0.0005	0.3576 ± 0.0017
PUFA–depleted	0.4994 ± 0.0002	0.4558 ± 0.0013

Table 6 — Average coarse–grained chain order parameter over all lipids obtained from MARTINI3 simulations of realistic and PUFA–depleted membranes, with and without CP55,940.

changes may modulate not only the structural response of the bilayer, but also the effective availability of the ligand within the membrane.

5. Discussion

5.1. Lipid composition as a modulator of ligand-membrane coupling

From the results, it appears that interactions between CP55,940 and the membrane depend strongly on the physical state and composition of the bilayer. In essence, the ligand induces structural changes within the membrane, while membrane — in turn — shapes the dynamic landscape experienced by the drug.

Considering both sides of the coupling, it is first worth noting that the membrane is a suitable environment for the drug. Its lipid–derived nature makes it easy to enter the bilayer, where it positions itself below the polar interface formed by the lipids. Consequently, the behavior of the ligand becomes strongly dependent on the nature of the surrounding lipid species. In a fluid environment, CP55,940 is free to arrange itself according to its own physicochemical preferences, keeping its polar groups close to the interface while the rest of the molecule explores a wide range of conformations. In a more constrained membrane, such freedom is considerably reduced. The ligand is no longer able to fully satisfy these preferences, and its insertion state becomes less clearly defined. What emerges is a competition between what the molecular architecture of the ligand tends to favor and what the membrane environment allows.

From the membrane perspective, it is equally clear that its physical state strongly influences the way it accommodates the ligand. The structural response observed upon CP55,940 insertion is not identical in fluid and ordered environments. What nevertheless appears common to all systems is that CP55,940 behaves somewhat like the small spacers used when laying tiles. By forcing a slight separation between neighboring lipids, it creates additional lateral free volume and facilitates hydration of the polar interface.

But what happens beneath the interface? Here, the answer depends strongly on the initial membrane state. In ordered membranes, CP55,940 disrupts the highly constrained packing of the acyl chains, partially restoring conformational freedom to the surrounding lipids. In that sense, one may be tempted to draw a parallel with the role of cholesterol, insofar as the ligand seems to shift the membrane away from an extreme gel–like organization toward a state displaying some characteristics of a liquid–ordered phase. The comparison, however, should stop there. Unlike cholesterol, in membranes whose hydrophobic core is already free to fluctuate, its presence has little to no impact on chain ordering. The ligand is simply not a sufficiently strong constraint to suppress these fluctuations, whereas cholesterol is precisely known for its ability to regulate them.

One encouraging aspect of the present study is that similar trends are observed across multiple levels of resolution. What is observed at the atomistic scale is somehow recovered within the coarse–grained description. Although some of the results remain preliminary, the overall picture consistently supports the idea that ligand–membrane coupling may contribute to drug efficacy, without implying that it is its sole determinant.

For lipophilic molecules, such as many compounds targeting the endocannabinoid system, one is naturally led to question what becomes of the ligand once it enters the membrane. The bilayer may first act as a reservoir, a pocket or even a trap, simply because it is capable of accommodating such molecules and retaining them for extended periods of time. Yet the membrane may play a more active role than that. It can also be viewed as a two–dimensional medium through which the ligand travels. In this picture the bilayer behaves almost like a waveguide, directing the motion of membrane–associated ligands, while its transport properties are themselves determined by lipid composition and state — a compact and ordered membrane does not offer the same landscape as a fluid and highly polyunsaturated one, as shown in the study.

This perspective may prove relevant for drug design. Beyond the classical question of receptor affinity, one may ask whether the affinity of a ligand for particular membrane environments, together with the mechanical properties of the membranes surrounding therapeutic targets, contributes to shaping pharmacological responses. If membrane composition alters ligand partitioning, mobility or accessibility to the receptor, then part of the dose–response relationship may already be determined before the first ligand–receptor contact occurs. While such ideas remain speculative at present, the results reported here suggest that they deserve further investigation.

5.2. Implications for lipidomic alterations

Before discussing the implications of PUFA depletion itself, it is worth realizing how fortunate we are to have access to such detailed lipidomics data in a biological environment where CB1 is highly expressed. The combination of modern lipidomics, molecular dynamics simulations and current computational resources now allows us to approach topical medical questions from a molecular perspective.

This was precisely the spirit of the present work which aimed to reproduce — at least partially — the complexity of the synaptic environment, in terms of lipids. We went further by proposing a deliberately PUFA–depleted model to explore a hypothesis currently receiving attention within the psychiatric community, namely the possible role of ω–3 and ω–6 deficiencies associated with several psychiatric disorders. Lipidomic alterations are measured in patients. Molecular dynamics cannot explain these observations on its own, nor establish causal relationships. What it can do, however, is provide a physical framework in which the consequences of such alterations can be explored, offering a bridge between clinical observations and molecular mechanisms.

From this bridge, several consequences can be envisioned. First, the healthy synaptic membrane considered here is particularly rich in PUFA lipids, a sign of high fluidity. From the perspective of the ligand — and perhaps even more so for endogenous ligands such as 2–AG — this means an environment that allows these molecules to fully exploit their own physicochemical nature. Membrane insertion can occur with limited perturbation of the surrounding lipids, and once inserted, the ligand remains free to diffuse, reorient and explore the bilayer. Such a membrane may therefore constitute a favorable medium for associated signaling processes.

The situation may differ in a more ordered or pathological membrane. Increased packing and reduced fluidity could make insertion less favorable and slow down membrane dynamics, but PUFA depletion may alter more than membrane fluidity alone. It also partially erodes the asymmetry that characterizes synaptic membranes. Lipids of different shapes are known to promote distinct curvature tendencies and stress states within the bilayer. Consequently, modifying their abundance is expected to alter, at least locally, the effective geometry of the membrane. Changes in geometry naturally come with changes in packing constraints, which may in turn affect ligand–membrane interactions. One may further speculate that such membranes become more selective. Different compounds may not be affected equally by lipidomic alterations. In that sense, the membrane could act not only as a transport medium, but also as a form of filter that favors some molecules over others. While this idea remains speculative and would require dedicated investigations, it naturally emerges from the picture suggested by the present results.

5.3. Methodological strengths and limitations

If this work has a particular strength, it is first its ability to explore several levels of description that ultimately tell a coherent story. Observing similar trends across atomistic and coarse–grained resolutions provides encouraging evidence for the robustness of the conclusions. Bridging scales is rarely straightforward, especially in membrane systems where different resolutions do not necessarily preserve the same observables. This multiscale approach therefore offers a valuable framework, capable of connecting fine molecular descriptions to larger biological contexts. Nevertheless, several limitations must be acknowledged.

The first is the absence of the receptor itself. Ultimately, the system of interest is the receptor–membrane–ligand triad, whereas the present work is only a slice of the complete picture. As a consequence, no direct conclusions can be drawn regarding receptor activation or pharmacological outcomes.

A second limitation concerns simulation timescales, particularly for the coarse–grained systems. Although the atomistic simulations extend over hundreds of nanoseconds, and the coarse–grained trajectories over several microseconds, these times remain insufficient to fully explore collective reorganizations. Processes such as lateral demixing, domain formation or preferential lipid recruitment around the ligand may occur on considerably longer timescales. Such phenomena could not only refine the interpretation of the results, but also provide valuable insight into the type of membrane environment preferentially occupied by CP55,940. Does the ligand preferentially partition into more fluid regions? Does it destabilize existing domains, or conversely contribute to their stabilization? These questions remain open.

Although CP55,940 constitutes a convenient molecular probe, cannabinoid ligands span a wide range of physicochemical properties. It therefore remains unclear to what extent the trends identified here can be generalized to endogenous ligands or to other synthetic compounds. Another limitation concerns the choice of a strongly depleted membrane. Because detailed lipidomic data for pathological synaptic membranes remain scarce, the present model should not be interpreted as a faithful reconstruction of any specific disease state. It only maximizes the physical consequences of PUFA depletion. While such an approach is useful for identifying possible mechanisms, the medical relevance of the conclusions must naturally be interpreted with caution and modulated in light of the severity of the depletion considered here.

No independent replicas were performed. Instead, computational resources were invested in extending the duration of individual trajectories, motivated by the relatively slow relaxation dynamics of membrane systems. The overall consistency observed across multiple observables provides confidence in the trends reported here. Nevertheless, future studies would greatly benefit from combining long trajectories with independent replicas in order to strengthen the statistical significance of the results.

A final limitation concerns experimental validation. This work was conducted alongside experimental investigations performed by Prof. Isabel Alves, Adeline Cœugnet and collaborators at the institute of Chemistry & Biology of Membranes & Nano–objects (CBMN) in Bordeaux. Using fluorescence probes in lipid vesicles, their work explores several structural properties that are conceptually related to those investigated in our atomistic membrane models. While some of the observed trends appear qualitatively consistent, the two approaches operate at different levels of description and cannot be directly compared quantitatively. The mechanisms proposed here remain primarily simulation–based and should not be regarded as experimentally validated. This limitation becomes even more pronounced for the realistic lipidomic membranes, for which no direct experimental counterpart currently exists.

Taken together, these limitations do not challenge the main trends identified in this work, but rather define the level at which the conclusions should be interpreted.

5.4. Perspectives

The results obtained in this work naturally open several perspectives for future investigations.

The most immediate extension consists in introducing the cannabinoid receptor itself into the simulations. While the present study focused on the membrane environment and on the physicochemical processes preceding receptor recognition, the next step is to determine how these membrane effects propagate to ligand access and receptor activation. In particular, it would be interesting to characterize the lateral entry pathways of cannabinoid ligands toward CB1 and to evaluate whether lipidomic alterations modify the probability of receptor encounter, the preferred access routes, … Extending these investigations to CB2 would provide an additional opportunity to compare two closely related receptors embedded in similar but distinct membrane environments.

The membrane models themselves can also be refined. Although the realistic membrane developed in this work represents a substantial step beyond simplified binary mixtures, it remains a reduced representation of the synaptic lipidome. Future simulations could incorporate a larger fraction of the experimentally identified lipid species and explore additional pathological lipidomic signatures. In particular, larger membrane patches and longer simulation times would provide access to collective phenomena such as lipid domain formation, lateral segregation and the emergence of preferential lipid environments surrounding inserted ligands or membrane proteins.

Beyond the objectives of the present study and the efforts aimed at understanding how membrane composition influences cannabinoid signaling, exploring this idea in greater detail and for a broader class of pharmacological targets may ultimately contribute to a more integrated description of pharmacological outcomes in physiological and pathological contexts.

As a final word, the collaboration with the group of Professor Isabel Alves at CBMN is currently giving rise to the preparation of a scientific article based on the combined experimental and computational results. This forthcoming work will constitute a natural continuation of the present study.

6. Conclusion

This work investigated how membrane composition modulates the behavior of the cannabinoid agonist CP55,940 before explicit receptor binding. The central idea was to consider the membrane not as a passive hydrophobic solvent, but as a structured and composition–dependent environment capable of shaping ligand insertion, orientation and availability.

At the atomistic level, CP55,940 was shown to partition favorably into lipid bilayers and to adopt an interfacial insertion mode in fluid DOPC membranes. This was confirmed by PMF profiles, showing a pronounced minimum inside the bilayer. In more ordered DPPC–based membranes, the ligand displays broader insertion and orientational distributions, suggesting that lipid packing constraints influence the accessible configurations of the molecule.

The membrane response to ligand insertion was also composition–dependent. CP55,940 generally increased the area per lipid, stated as a depacking effect produced by the insertion of a small hydrophobic molecule within the bilayer. Changes in chain order and hydration further indicated that the ligand perturbs membrane organization, particularly in initially ordered systems. These results support a bidirectional view of ligand–membrane coupling: the membrane constrains the ligand, while the ligand modifies the membrane.

The coarse–grained MARTINI3 simulations extended this picture to larger and more realistic lipidomic–inspired membrane models. The comparison between the realistic synaptic membrane and its PUFA–depleted counterpart showed that removal of polyunsaturated chains leads to a more compact and more ordered bilayer. In contrast, CP55,940 partially counteracts this organization by increasing the APL and decreasing the average chain order parameter. Fewer ligands were inserted in the PUFA–depleted membrane on the simulated timescale, suggesting that PUFA depletion may reduce membrane permissiveness toward lipophilic cannabinoid ligands.

These results suggest that lipidomic alterations may influence cannabinoid signaling upstream of receptor binding, by reshaping the membrane environment through which ligands partition, diffuse and orient before reaching CB1. This interpretation remains preliminary, since the receptor itself was not included and longer simulations with independent replicas would be required to quantify lateral diffusion, lipid reorganization and equilibrium partitioning more rigorously. Nevertheless, the present work provides a physically coherent framework linking membrane composition, ligand insertion and membrane structural response.

More broadly, this study supports the relevance of lipidomics–informed molecular simulations for computational pharmacology. By combining atomistic resolution with coarse–grained realistic membrane models, it becomes possible to connect molecular mechanisms with biologically motivated lipid compositions. Future work including CB1, CB2, additional ligands and experimental validation will be necessary to determine how these membrane–mediated effects contribute to cannabinoid receptor functions.

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